Maximum marks for the paper (Excluding bonus question)= 70

Semester	IV
Paper Code	**CS0222**
Paper Title	ADVANCED OPERATING SYSTEM
Number of teaching hours per week	4
Total number of teaching hours per semester	60
Number of credits	4

Course Objectives

- To make the student understand with various function of Operating system.
- The knowledge of resources management of Operating system.
- The knowledge about various problems and solution is distributed system.
- The knowledge about fault tolerance.

UNIT I **(12)**

INTRODUCTION

Overview - Functions of an Operating System – Design Approaches – Types of Advanced Operating System - Synchronization Mechanisms – Concept of a Process, Concurrent Processes – The Critical Section Problem, Other Synchronization Problems – Language Mechanisms for Synchronization – Axiomatic Verification of Parallel Programs - Process Deadlocks - Preliminaries – Models of Deadlocks, Resources, System State – Necessary and Sufficient conditions for a Deadlock – Systems with Single-Unit Requests, Consumable Resources, and Reusable Resources.

UNIT II **(12)**

DISTRIBUTED OPERATING SYSTEMS

Introduction – Issues – Communication network and Primitives, – Theoretical Foundations: Inherent Limitations - Lamport's Logical Clock; Vector Clock; Causal Ordering; Global State; Cuts; Termination Detection. Distributed Mutual Exclusion Non-Token Based Algorithms – Lamport's Algorithm - Token-Based Algorithms Suzuki-Kasami's Broadcast Algorithm – Distributed Deadlock Detection – Issues Centralized Deadlock-Detection Algorithms - Distributed Deadlock-Detection Algorithms. Agreement Protocols – Classification(two)- Solutions(two) –Applications.

UNIT III **(12)**

DISTRIBUTED RESOURCE MANAGEMENT

Distributed File systems – Architecture – Mechanisms – Design Issues – Distributed Shared Memory – Architecture – Algorithm(two) – Protocols - Design Issues. Distributed Scheduling – Issues – Components(two) – Algorithms (two).

UNIT IV **(12)**

FAILURE RECOVERY AND FAULT TOLERANCE

Basic Concepts-Classification of Failures – Basic Approaches to Recovery; Recovery in Concurrent System; Synchronous and Asynchronous Check pointing and Recovery; Check pointing in Distributed Database Systems; Fault Tolerance; Issues - Two-phase and Nonblocking Commit Protocols; Voting Protocols; Dynamic Voting Protocols.

UNIT V **(12)**

MULTIPROCESSOR AND DATABASE OPERATING SYSTEMS

Structures – Design Issues – Threads – Process Synchronization – Processor Scheduling – Memory Management – Reliability / Fault Tolerance(two); Database Operating Systems – Introduction – Concurrency Control – Distributed Database Systems – Concurrency Control Algorithms(two).

Self Study – Analysis of any open source operating system and study of algorithms in the chapters.

REFERENCES

- Andrew S. Tanenbaum, "Modern Operating Systems", Second Edition, Addison Wesley, 2001.
- Mukesh Singhal and N. G. Shivaratri, "Advanced Concepts in Operating Systems", McGraw- Hill, 2001
- Abraham Silberschatz, Peter B. Galvin, G. Gagne, "Operating System Concepts", Sixth Edition, Addison Wesley Publishing Co., 2003.

BLUEPRINT

Code number: **CS0222**

Title of the paper: **ADVANCED OPERATING SYSTEM**

Chapter	Number of Hours	Total marks for which the questions are to be asked (including bonus questions)